Redesigning Gynecologic Care: A Soft Robotic Solution to Painful Cervical Stabilization

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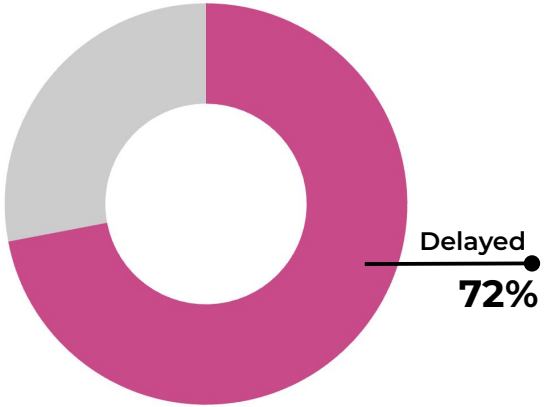
Category: Biosensors and Instrumentation

AGENDA

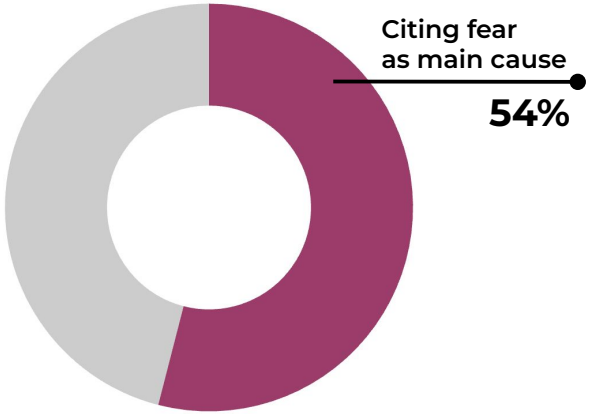
1. Challenge & Status Quo
2. Diversity Considerations
3. Ideation Phase Work
4. Development Phase Timeline

Fear of pain, discomfort, and invasive tools is one of the leading reasons women avoid or delay essential gynecologic care.

Percent of Women in US Who Delay Their Gynecological Visits



Percent of Women in US Who Report Fear as Their Delay Cause



60–70% of women in the UK and Sweden also delay gynecologic visits due to fear, embarrassment, or anticipated pain.

- Emotional barriers are noted as the **greatest threat** to cervical cancer screenings compared to time and money.
- Avoidance of gynecologic care due to fear leads to **late detection** of cervical cancer and other diseases.

Emotional barriers to gynecologic care are **unevenly distributed.**

Fear-driven avoidance **disproportionately affects** young adults, low-income patients, and marginalized communities, which **widens existing inequities** in reproductive healthcare access.

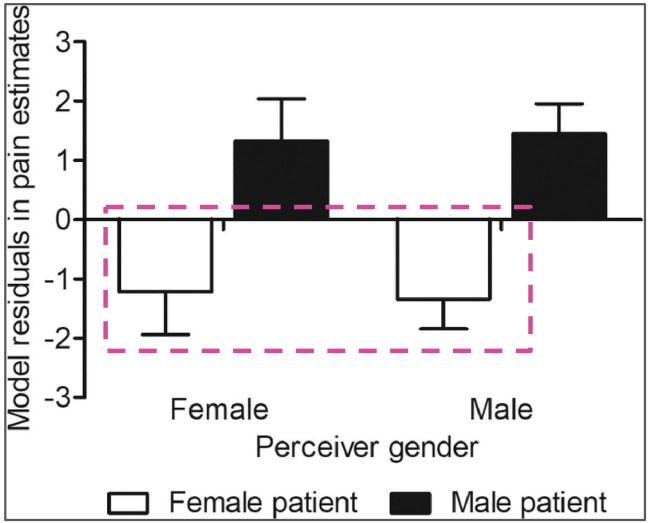
Sources: Survey Finds Women Are Skipping Their OB/GYN Exams, Increasing Risks for Cervical Cancer

Women's pain during procedures are underestimated in medical studies, which leads to this problem staying unaddressed.

In studies for pain measurement, **women's pain are underestimated** while men's are **overestimated** when they're experiencing equal amounts of pain.

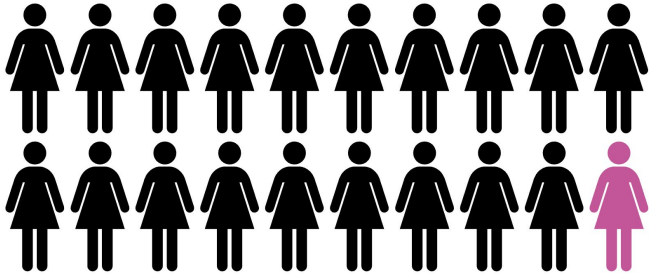
The power of this pain tolerance bias can be understood by seeing how **women observers also underestimate** the pain that women go through.

Gap in Perceived Pain Estimates at The Same Pain Level for Both Genders



This underestimation correlates with the lack of pain relievers provided for women.

Pain relievers (medication, local anesthesia) exist, but are not provided commonly



Only 1 in 20 women receive pain medication while getting an IUD.

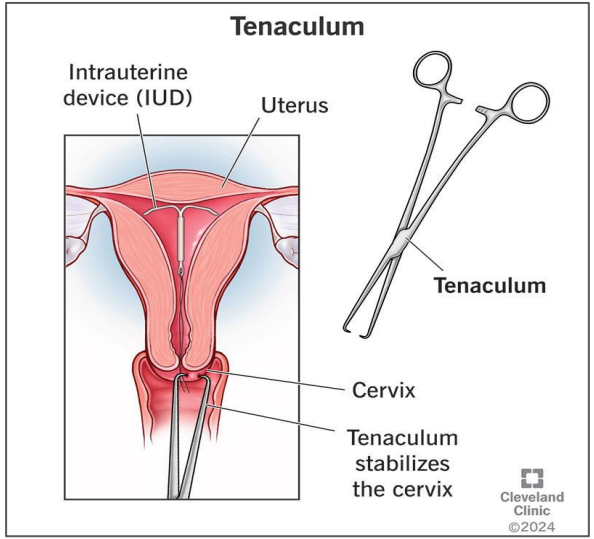
Sources: Gender Biases in Estimation of Others' Pain, Women's health: Pain medication remains rare for women receiving IUDs

IUD is one of the most effective contraceptive methods available, also one of the most painful: Tenaculum design is a major part of that pain.

For an IUD survey, 41.6% of IUD users reported their insertion pain was “unacceptable.”

Major reasons behind IUD pain are from uterine cramps due to the IUD itself and **tenaculum use.**

- A **tenaculum** is a hinged, pointed forceps used in gynecology to **secure the cervix**
- It penetrates the cervical tissue, providing traction for procedures such as **IUD insertion, biopsy**, and **hysteroscopy.**



The tenaculum is an example of medical devices that needs to be improved through design

The history of gynecological development lies in **exploitation of women’s bodies**, especially the bodies of enslaved Black women, and the **normalization of pain** as an acceptable cost of care.

The tenaculum traces its design to a 19th-century **bullet extractor**, a tool built for battlefield surgery, with virtually **no consideration for female anatomy** or patient comfort.

Sources: What is a tenaculum?, For Over 100 Years Women Have Endured Pain in Gynecology

Aim of the soft robotic tenaculum is to lower pain during gynecological procedures, while being a design that is anatomically inclusive

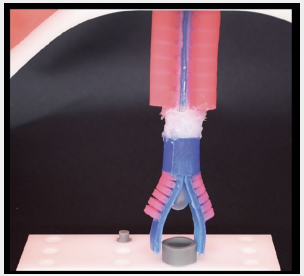
Research during my ideation phase showed key information regarding materials, possible gripper designs, and activation methods.

Material Selection

- For a medical device like tenaculum, its material needs to be highly biocompatible like **medical-grade silicone**.
- However, **DragonSkin silicone** is a safe alternative for the molding of soft robotic parts.
- We can also apply medical-grade silicone **coating** to the tip of the gripper.

Gripper Design Ideas

- The main cause of tenaculum design is the sharp needle-like tip of the device.
- A gripper that distributes the pain with a larger surface
- 3 degrees of freedom for movement



Activation Methods

- There are multiple methods available for powering a soft robot, each differing with power or cost.
- Motor usage can work, while having drawbacks due to cost and ease of use.
- Pneumatic actuators that work like a syringe can power the gripper with correct attachment & calculations.

Sources: A Soft Continuum Robot with Multi-Modal Shape and Contact Force Sensing for Minimally Invasive Surgery

Timeline for the development of this project

Phase 1
- Coming up with a design idea
- Developing the rationale behind the design

Phase 2
- Selecting & ordering materials
- Starting literature review
- Getting in contact with Material Robotics Lab

Phase 3
- Finishing literature review on all topics
- Starting CAD modeling for soft robot mold
- Deciding on an activation method

Phase 4
- 3D print the mold design and fabricate the silicone model
- Test out the model & redesign if necessary

Phase 5
- Perform mechanical testing
- Evaluate pressure distribution on cervical-tissue analogs

Phase 6
- Gather feedback on device shape, handle design, and performance
- Assess ergonomics and ease of use

Phase 7
- Build a cost and manufacturing feasibility overview
- Plan for adoption or commercialization

Phase 8
- Produce final prototype with incorporated usability & design changes
- Final presentation